How Much Targeted Training Does It Take to Top a Multi-Task Leaderboard? Minimum Investment Under Different Aggregation Rules

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Abstract

A leaderboard turns per-task scores into one rank, so a developer can try to lift a chosen model by training on data related to individual sub-datasets. How much such targeted training is needed to put the model first, and how does that depend on the aggregation rule? We define the investment as the minimum raise of the target's task scores, within a per-task cap, that makes it strictly first, and measure it in points, tasks touched and largest single-task jump. We solve it exactly for the mean, median, minimum, Borda and Condorcet-winner rules (and Copeland on HELM), prove closed forms for the first three, and apply it to every model of the Open LLM Leaderboard (4495 targets, 6 tasks) and of HELM MMLU (78 targets, 57 subjects). With caps taken from observed fine-tuning gains, first place is reachable for only 12% of OLLB targets under the mean (largest feasible rank 901) and 19% of HELM targets (largest feasible rank 16); with a 100-point cap every target can reach it. Among feasible targets the mean is the costliest rule and the median and minimum the cheapest: at OLLB ranks 11–50 the other rules need 0.18–0.50 of the mean's total points. The ordering of rules by feasibility changes with the cap, and the conclusions hold for fixed rivals and score tables only; the HELM cap is transferred from OLLB and decides its frontier (98.7% feasible with a loose pooled cap).

1 Introduction

Multi-task benchmarks rank models by aggregating per-task scores into one number, and the resulting rank influences which models are adopted, published and funded. A developer therefore has an incentive to improve a chosen target model specifically on the benchmark's sub-datasets, for instance by training on data related to them. How much such targeted improvement is needed to reach first place, and how that amount depends on the aggregation rule, is the question we study on public leaderboard data.

Our approach is deliberately simple and exact. We take the score table of a public leaderboard as fixed, let the target raise its own scores task by task up to a cap, and compute the *minimum* raise that makes it strictly first under a given aggregation rule. The cap is the central modelling assumption: it stands for how far training on related data can move one score. We therefore report results under a data-grounded cap and an oracle cap, and report infeasibility (no allowed raise makes the target first) as an outcome rather than a missing value.

Contributions. (i) A formulation of the minimum targeted investment with three cost measures (L_1, K, L_∞) and an explicit family of caps (Section 3). (ii) Exact solvers for score rules (mean,

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median, minimum, power and trimmed means) and rank rules (Borda, Condorcet winner, plurality, Copeland), tested against brute force, with closed forms and proofs for the mean, minimum and median (Section 3.4, Appendix A). (iii) A study of every model on two public leaderboards, with bootstrap and evaluation-noise intervals (Sections 4–5).

Findings. Under caps grounded in observed fine-tuning gains, first place is out of reach for most models under the score rules on both datasets and under Borda, the Condorcet requirement and Copeland on HELM (rank rules on OLLB are solved for the 50 best targets only): the feasible fraction under the mean is 12% on OLLB and 19% on HELM, and the rule changes how far down the board it is reachable. Where it is reachable, the mean rule costs the most points, and order-statistic and rank rules cost between 18% and 50% of that at OLLB ranks 11–50. Which rule is easiest to top by feasibility depends on the cap and the dataset, so no rule is uniformly the most exposed; the cost ordering is more stable. Of three hypotheses fixed beforehand, one held, one held in part and one was not supported (Section 5). All conclusions are conditional on the cap and on the other models being fixed; Section 6 lists where they fail.

2 Related work

Benchmarks and leaderboards. MMLU [Hendrycks et al., 2020] aggregates 57 subjects, and HELM [Liang et al., 2022] releases per-scenario scores and raw model outputs for many models; BIG-Bench Hard [Suzgun et al., 2022] selects hard tasks from BIG-Bench. These provide the multi-task score matrices that we analyse.

Fragility of rankings. Dehghani et al. [2021] show that the relative order of methods can change with the choice of benchmark tasks (the "benchmark lottery"). Singh et al. [2025] document how private testing and unequal data access distort the Chatbot Arena leaderboard. Both concern what is measured and disclosed; we instead hold the leaderboard fixed and ask what a single developer must change for one model to win.

Aggregation and social choice. Lanctot et al. [2023] treat each task as a voter and use voting theory to build cross-task evaluation; the Handbook of Computational Social Choice [Brandt et al., 2016] covers the rules (Borda, Condorcet-consistent rules, plurality) we reuse. Our rules are the same families, but our question is the cost of changing the winner, not the choice of rule.

Training on the test task and contamination. Zhou et al. [2023] show that benchmark leakage into training can raise scores sharply, and Yang et al. [2023] show that rephrased test samples evade common decontamination. Chen et al. [2026] analyse, in a game-theoretic model, the incentives that benchmarks create for strategic post-training. Our analysis is empirical and combinatorial: given a fixed score table, we compute the exact minimum targeted improvement per rule, under explicit assumptions about how much any one task can be improved.

Positioning. Closest in question is Chen et al. [2026], who model developers choosing how much to spend on benchmark-specific improvement of a single benchmark score, prove non-existence of equilibria for current protocols and a unique equilibrium for tune-before-test. That analysis concerns one score per model; the choice of aggregation rule over tasks does not enter it. We take the opposite stance: no game and no equilibrium, a fixed table, and the exact cost for one developer under each of several aggregation rules. Colombo et al. [2022] propose a social-choice-based aggregation for NLP benchmarks and show it changes conclusions relative to the mean; we do not propose a rule but measure how costly it is to move a model to first place under each. The computational-social-choice literature studies the analogous *bribery* problem, the cost of changing votes so that a chosen candidate wins under a voting rule [Faliszewski et al., 2009, Brandt et al., 2016]. Our setting differs in that the manipulator changes its own scores on tasks (the "voters") and pays in score points under per-task caps, which gives closed forms for score rules and integer programs for rank rules.

3 Problem setup

3.1 Leaderboard, target and investment

A leaderboard is a score matrix $X \in [0,100]^{M\times n}$ of M models on n tasks. Fix a target model i with score vector $x = X_{i\cdot}$; all other rows stay fixed. In targeted training the developer can raise the target's score on task t to any $y_t \in [x_t, c_t]$, where c_t is a *cap* (Section 3.2). An aggregation rule f maps the score vectors to a ranking; the target is *first* if it is strictly above every other model under f (by a margin $\varepsilon = 10^{-3}$ points for score rules). We report the investment in three ways:

$$L_1 = \sum_t (y_t - x_t), \qquad K = \#\{t : y_t > x_t\}, \qquad L_\infty = \max_t (y_t - x_t),$$

the total points, the number of tasks touched, and the largest single-task jump. The minimum of each is taken separately; L_∞ is also the cost of the uniform "general improvement" that raises every task by the same amount. For comparison across rules and datasets we also report K/n and L_1/n.

3.2 Caps: how much can one task be improved?

Without a bound on y_t the question is trivial (put every task at 100). The cap c_t is therefore the central modelling assumption, and we never collapse it to a single choice. Two caps appear in the main text: `ceil`, $c_t = 100$, an oracle upper bound on what can be done and so a lower bound on any required investment; and `emp99`, $c_t = \min(100, x_t + g_t^{99})$ where g_t^{99} is the 99th percentile of the observed gains (fine-tune minus its declared base model) on task t in the Open LLM Leaderboard. `emp99` is the only data-grounded cap. It inherits a selection bias, since only published fine-tunes appear, and it measures gains attained by the community, not the gains a determined developer could attain. Further caps (best existing score $+0$ or $+2$, and at most $+5/+10/+20$ points per task) are reported in the appendix. When no $y \le c$ makes the target first, the case is *infeasible*; infeasibility is reported as a rate and never coded as a large investment, and all investment statistics are conditional on feasibility.

3.3 Aggregation rules

Score rules apply a function f to the score vector of each model: the *mean* (the rule of both leaderboards we use), the *median*, and the *minimum* (worst-task score). Rank rules use only the order of models on each task: *Borda* (sum over tasks of the number of models beaten, ties counting one half; equivalent to mean rank) and the *Condorcet-winner* requirement (the target beats every rival on a strict majority of tasks, ties counting one half). Being a Condorcet winner is sufficient for being first under any Condorcet-consistent rule such as Copeland or Schulze, so its investment upper-bounds theirs. Geometric, harmonic and power means, the trimmed mean and plurality are in the appendix.

3.4 Exact minimum-investment solvers

For a score rule let $\tau = \max_{j\neq i} f(X_{j\cdot})$ be the level of the best rival; the target is first iff $f(y) \ge \tau + \varepsilon$. For the mean, the minimum and the median the minimum investment has a closed form (Propositions 1–3, proved in Appendix A); for power means and the trimmed mean we solve a convex program or enumerate exactly. For rank rules a raise of task t to level $s + \varepsilon$ passes exactly the models with score at most s on that task, so only these finitely many levels need be considered, and we solve the resulting integer program exactly with HiGHS, adding rival constraints lazily. Every solver is checked against brute-force search on small random instances (unit tests in the supplementary code). For each target we report L_1 and K, each minimised separately, and L_∞ for the uniform baseline.

Proposition 1 (Mean)**.** *Let f be the mean and $h_t = c_t - x_t$. The target can become first iff $\sum_t c_t > n\tau$ (up to ε). The minimum L_1 is $n(\tau - f(x))$ regardless of how it is spread, the minimum K is the least k for which the k largest h_t sum to at least this amount, and the minimum L_∞ is $\tau - f(x)$ (raise all tasks equally) when no cap binds.*

Proposition 2 (Minimum)**.** *Let $f = \min_t y_t$. The target can become first iff $c_t > \tau$ for all t with $x_t \le \tau$. Then $L_1 = \sum_t (\tau - x_t)^+$ and $K = \#\{t : x_t \le \tau\}$: every deficient task must be raised.*

Proposition 3 (Median, n odd)**.** *Let $m = (n+1)/2$, $d_t = (\tau - x_t)^+$ for tasks with $c_t > \tau$ and $d_t = \infty$ otherwise. The target is first iff at least m tasks end above τ. The minimum L_1 is the sum of the $(m - \#\{t : x_t > \tau\})^+$ smallest finite d_t among tasks with $x_t \le \tau$, and K equals that count; the target can be made first iff enough finite d_t exist.*

(All statements are for $\varepsilon \to 0$; the code uses $\varepsilon = 10^{-3}$ points. For even n the median is the average of the two middle order statistics and we only have the computed solution.) We verify Propositions 1–3 numerically against the solvers (Section 4); no theorem is claimed for the rank rules.

4 Experimental setup

Data. We use two public leaderboards, each at a fixed, recorded version (checksums in the supplementary reproduction notes). *HELM MMLU*, release v1.13.0 (summary dated 2025-01-10): the exact-match score of 79 models on each of the 57 MMLU subjects (79×57, no missing cell). *Open LLM Leaderboard* (OLLB) `contents`, dataset commit `9c09a7c` (2025-03-20): 4576 rows, of which 4497 have distinct model names after we keep the best-scoring variant of each duplicated name, and 4496 after removing one flagged model; each model has six normalised task scores (IFEval, BBH, MATH Lvl 5, GPQA, MUSR, MMLU-PRO), so $n = 6$. Merged models are kept in the main analysis and dropped in a robustness slice. The mean we use equals the sites' own aggregate (OLLB `Average`, HELM "MMLU All Subjects") to within 6×10^{-14} points on every model. Neither source declares a licence for the score data; we redistribute only the derived score tables.

Targets, caps and what is reported. Every model other than the current leader is a target in turn, with all others fixed (4495 targets on OLLB, 78 on HELM). On OLLB the primary cap is `emp99` (per-task 99th percentile of the gain of a fine-tune over its declared base model, estimated from 540 such pairs in the same leaderboard). HELM has no base/fine-tune pairs, so its primary cap, `emp99T`, applies the OLLB MMLU-PRO 99th-percentile gain to all 57 subjects; this is a transfer assumption, not a measurement on HELM. The oracle cap `ceil` is reported beside it. All investment statistics are medians over feasible targets, shown together with the feasible fraction. OLLB rank rules are exact integer programs, which at $M = 4496$ models we run only for the 50 best targets under the mean; we apply a 120 s limit per instance and report time-outs as censored, never as infeasible. A slice of the 499 best OLLB targets is solved for the ordinal rules as a check (Appendix C), and Copeland is solved exactly on HELM only. The robustness runs solve Borda and Condorcet for seven chosen OLLB targets of ranks 2–100, beyond the 50 best.

5 Results

All statements below are about targets that can be made first within the cap; the feasible fraction is shown beside every cost, and a rule is called cheaper than another only on targets that are feasible under both.

The closed forms are exact (Q1). Propositions 1–3 agree with the exact solvers on every target of both datasets and every cap: the largest deviation in L_1 is below 2×10^{-12} points, the deviation in K is zero, and no feasibility verdict differs (Appendix Table 3). The OLLB median is solver-only because $n = 6$ is even.

Feasibility depends on rank and on the rule (Q3). Under the data-grounded caps, targeted training cannot make most models first (Figure 1, left). On OLLB the feasible fraction under the mean is 1.00 up to rank 300 and 0.55 up to rank 1000, whereas under the median it is 0.91 up to rank 300 and 0.45 up to rank 1000 (every target is feasible only up to rank 394 under the mean and 90 under the median); no target beyond rank 901 (mean) or 1207 (median) is feasible; on HELM every target at ranks 2–10 is feasible under the mean, median, Borda and Condorcet rules, and the fraction falls steeply afterwards (Table 2: for ranks 11–50 only 6 of 40 targets are feasible under the mean and 17 of 40 under the median). The minimum rule is the most restrictive on both datasets, since every task below the best rival's weakest score must be raised and any one cap can block it (Proposition 2). The cap, not the rule, decides feasibility far from the top: with the oracle cap `ceil` every target is feasible (Appendix Figure 4).

Table 1 gives the frontier. On OLLB the fraction of targets that can reach first place under `emp99` is 12% under the mean, 10% under the median and 6% under the minimum; every target is feasible only up to rank 394 (mean), 90 (median) and 5 (minimum), and the largest feasible rank is 901, 1207 and 2225. On HELM under `emp99T` the fractions are 19%, 33% and 12%, every target is feasible up to rank 16, 19 and 6, and the largest feasible rank is 16, 29 and 20; the Condorcet requirement reaches

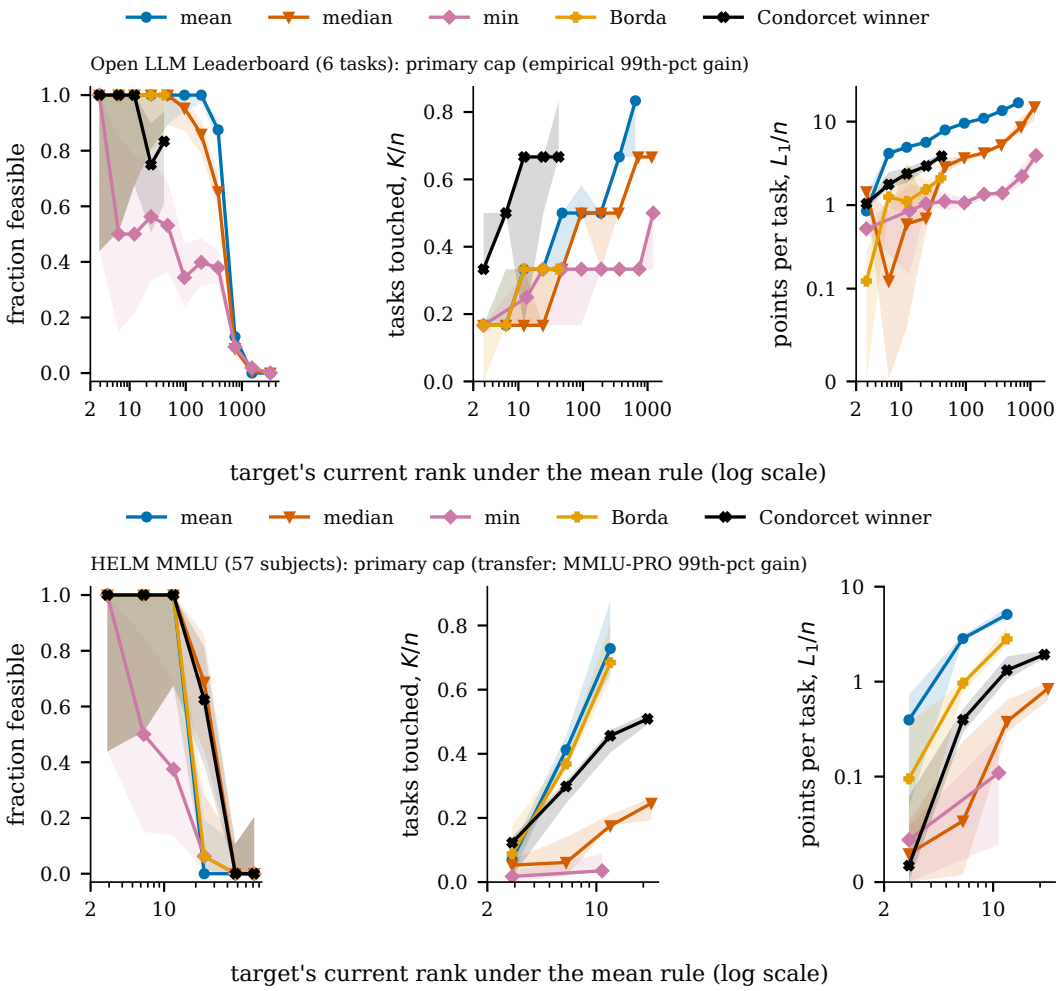


Figure 1: Minimum targeted investment to become first versus the target's current rank under the mean, primary caps. Left: fraction of targets that can reach first at all. Middle and right: tasks touched (K/n) and total points per task (L_1/n), *among feasible targets*. Top: OLLB, $n = 6$, cap `emp99` (4495 targets; Borda and Condorcet only for the 50 best). Bottom: HELM MMLU, $n = 57$, cap `emp99T` (78 targets; the cap is transferred from OLLB). Points are bin medians (bins double from rank 2; bins with fewer than 3 targets omitted), bands are 95% bootstrap intervals over targets. Generated by `figures/src/fig_rank_rules.py`.

rank 26, Copeland (solved exactly) the same 25 targets and rank 26, and Borda 17. Which rule is the most exposed therefore depends on the dataset and on the cap. On HELM the median is feasible for more targets than the mean at every cap of the sweep up to $+20$ points per task (21% against 14% at $+5$, 51% against 37% at $+10$, 85% against 72% at $+20$, 33% against 19% under `emp99T`). On OLLB the mean is feasible for more targets than the median under `emp99` (12% against 10%) and at $+20$ points (18% against 14%), whereas the median and the minimum reach isolated targets much further down (largest feasible rank 1207 and 2225 against 901). Among the rules of the main text the minimum is the most restrictive on HELM under every capped scenario and on OLLB under `emp99`; under the uniform caps of $+5$, $+10$ and $+20$ points on OLLB it is feasible for 2%, 10% and 64% of targets against 0.3%, 3% and 18% for the mean, because it only requires the weakest tasks to be lifted. The HELM headline is set by the transfer assumption: with the pooled six-task cap `emp99P` (the 99th percentile of gains pooled over the six OLLB tasks, $+44.9$ points on every task) 77 of 78 HELM targets (98.7%) are feasible under the mean, median and minimum rules, against 19% under `emp99T` ($+7.06$ points per subject), and the sweep gives 14%, 37% and 72% for the mean at $+5$,

Table 1: Feasibility frontier under the primary cap (`emp99` on OLLB, `emp99T` on HELM) and the oracle cap (`ceil`). *share*: fraction of targets for which first place is reachable (Wilson 95% intervals); r^*: largest rank under the mean among feasible targets (may be an isolated case); *all feas. up to*: largest r such that every target of rank $\leq r$ is feasible; *first bin < 1/2*: first rank bin (bins double from rank 2) in which fewer than half are feasible. †: the 50 best targets under the mean only, not comparable with the score rules' fractions. Generated by `figures/src/tables.py`.

cap	rule	targets	feasible	share [95% CI]	r^*	all feas. up to	first bin < 1/2
primary (`emp99`)	mean	4495	546	0.12 [0.11, 0.13]	901	394	513–1024
primary (`emp99`)	median	4495	451	0.10 [0.09, 0.11]	1207	90	513–1024
primary (`emp99`)	min	4495	273	0.06 [0.05, 0.07]	2225	5	65–128
primary (`emp99`)	Borda†	49	49	1.00 [0.93, 1.00]	50	50	none
primary (`emp99`)	Condorcet†	49	42	0.86 [0.73, 0.93]	50	18	none
oracle (`ceil`)	mean	4495	4495	1.00 [1.00, 1.00]	4496	4496	none
oracle (`ceil`)	median	4495	4495	1.00 [1.00, 1.00]	4496	4496	none
oracle (`ceil`)	min	4495	4495	1.00 [1.00, 1.00]	4496	4496	none
oracle (`ceil`)	Borda†	49	49	1.00 [0.93, 1.00]	50	50	none
oracle (`ceil`)	Condorcet†	49	49	1.00 [0.93, 1.00]	50	50	none

cap	rule	targets	feasible	share [95% CI]	r^*	all feas. up to	first bin < 1/2
primary (`emp99T`)	mean	78	15	0.19 [0.12, 0.29]	16	16	17–32
primary (`emp99T`)	median	78	26	0.33 [0.24, 0.44]	29	19	33–64
primary (`emp99T`)	min	78	9	0.12 [0.06, 0.20]	20	6	9–16
primary (`emp99T`)	Borda	78	16	0.21 [0.13, 0.31]	17	17	17–32
primary (`emp99T`)	Condorcet	78	25	0.32 [0.23, 0.43]	26	26	33–64
primary (`emp99T`)	Copeland	78	25	0.32 [0.23, 0.43]	26	26	33–64
oracle (`ceil`)	mean	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	median	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	min	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	Borda	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	Condorcet	78	78	1.00 [0.95, 1.00]	79	79	none
oracle (`ceil`)	Copeland	78	78	1.00 [0.95, 1.00]	79	79	none

$+10$ and $+20$ points. The largest feasible rank can be an isolated case (OLLB, minimum), which is why the column "all feasible up to" is shown.

Cost, conditional on feasibility, differs strongly by rule (Q2). Among feasible targets, the mean rule is the most expensive on both datasets and the minimum and median rules the cheapest (Figure 1, right; Table 2). On OLLB at ranks 11–50 the median feasible target needs 37.8 points in total under the mean, 9.1 under the median, 6.9 under the minimum, 11.7 under Borda and 20.5 under the Condorcet-winner requirement; relative to the mean on jointly feasible targets, the ratios are 0.22, 0.18, 0.27 and 0.50. On HELM at ranks 11–50 the corresponding totals are 321, 38, 10, 188 and 105 points. The reason is structural: the mean must close the whole summed gap (n times the gap, Proposition 1), whereas order-statistic rules pay only for the cheapest majority of tasks or the weakest tasks. Rank rules sit in between, because they pay for passing rivals task by task. Over all jointly feasible targets (Appendix Table 5, with the numbers of jointly feasible targets) the median ratio of L_1 to the mean is 0.39 (median), 0.11 (minimum), 0.27 (Borda) and 0.52 (Condorcet) on OLLB, and 0.07, 0.05, 0.51 and 0.21 (Copeland 0.21) on HELM, each with a bootstrap interval in the table. Table 2 shows intervals; the ordering mean $>$ Condorcet $>$ Borda $\gtrsim$ median $\gtrsim$ min on OLLB is not uniform across rank bins (the intervals at ranks 2–10 are wide, with $n = 9$ targets).

Which of our predictions held. The plan fixed three hypotheses before the full results (H1–H3, Appendix D). H2 (cost rises with rank, feasibility falls as caps tighten) holds on both datasets. H1 (order-statistic and majority rules need *more* tasks but possibly fewer points than the mean) holds only in part: they need fewer points, but not more tasks, except that the Condorcet requirement touches more tasks than the mean on OLLB at $n = 6$ ($K = 4$ against 2 at ranks 11–50); on HELM the median needs fewer tasks than the mean ($K = 12$ against 46). H3 is not supported for the median (below).

Table 2: Median investment to become first among feasible targets, by rank stratum and rule, primary caps: OLLB (`emp99`, top) and HELM (`emp99T`, bottom). "feas." = feasible targets / targets in the stratum; K, L_1 with 95% bootstrap intervals over targets; the last column is the median ratio of L_1 to the mean rule over targets feasible under both. OLLB Borda and Condorcet use the 50 best targets only. Generated by `figures/src/tables.py`.

	rank 2–10				rank 11–50			
	feas.	K	L_1	L_1/L_1^{mean}	feas.	K	L_1	L_1/L_1^{mean}
mean	9/9	1.0 [1.0, 2.0]	24.6 [5.1, 27.7]	–	40/40	2.0 [2.0, 2.0]	37.8 [33.9, 41.0]	–
median	9/9	1.0 [1.0, 1.0]	0.8 [0.0, 8.4]	0.03 [0.00, 1.70] (9)	40/40	1.0 [1.0, 2.0]	9.1 [4.4, 14.6]	0.22 [0.13, 0.40] (40)
min	5/9	1.0 [1.0, 2.0]	3.7 [2.8, 14.6]	0.61 [0.23, 0.77] (5)	24/40	2.0 [1.5, 2.0]	6.9 [5.9, 8.4]	0.18 [0.14, 0.19] (24)
Borda	9/9	1.0 [1.0, 2.0]	5.6 [0.7, 6.2]	0.20 [0.09, 0.35] (9)	40/40	2.0 [2.0, 2.0]	11.7 [9.5, 12.6]	0.27 [0.25, 0.31] (40)
Condorcet	9/9	3.0 [2.0, 4.0]	10.0 [6.7, 16.7]	0.59 [0.41, 1.04] (9)	33/40	4.0 [4.0, 4.0]	20.5 [18.0, 22.4]	0.50 [0.47, 0.55] (33)

	rank 2–10				rank 11–50			
	feas.	K	L_1	L_1/L_1^{mean}	feas.	K	L_1	L_1/L_1^{mean}
mean	9/9	23.0 [4.0, 32.0]	156.3 [22.5, 222.4]	–	6/40	46.0 [40.0, 51.0]	321.2 [278.2, 351.3]	–
median	9/9	4.0 [1.0, 10.0]	3.3 [0.4, 16.6]	0.04 [0.00, 0.09] (9)	17/40	12.0 [10.0, 15.0]	37.5 [31.7, 53.6]	0.09 [0.07, 0.11] (6)
min	6/9	1.5 [0.5, 2.5]	2.3 [1.0, 5.3]	0.03 [0.00, 0.09] (6)	3/40	3.0 [2.0, 5.0]	10.4 [6.2, 15.5]	0.04 [0.02, 0.06] (2)
Borda	9/9	20.0 [5.0, 30.0]	54.0 [5.6, 100.5]	0.36 [0.28, 0.94] (9)	7/40	45.0 [38.0, 48.0]	187.8 [143.1, 230.8]	0.55 [0.51, 0.64] (6)
Condorcet	9/9	16.0 [7.0, 22.0]	20.7 [0.9, 57.6]	0.16 [0.02, 0.22] (9)	16/40	28.0 [27.0, 29.0]	104.6 [90.4, 113.9]	0.28 [0.21, 0.31] (6)
Copeland	9/9	16.0 [7.0, 22.0]	20.7 [0.9, 57.6]	0.16 [0.02, 0.22] (9)	16/40	28.0 [27.0, 29.0]	104.6 [90.4, 113.9]	0.28 [0.21, 0.31] (6)

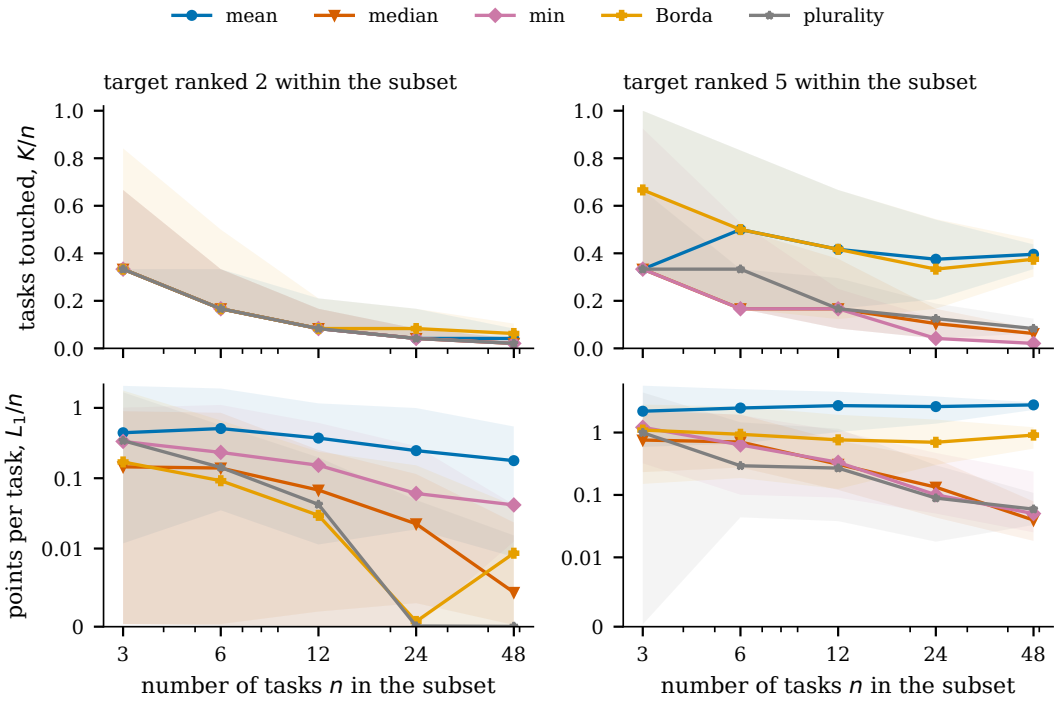


Figure 2: HELM: investment versus the number of tasks n in a random subset of subjects, for the target ranked second (left) or fifth (right) within the subset under the mean; bands span the 2.5–97.5 percentiles over 60 subset draws, among feasible draws, cap `emp99T`. Generated by `figures/src/fig_scaling.py`.

Number of tasks (Q4). Figure 2 draws random subsets of $n \in \{3, 6, 12, 24, 48\}$ HELM subjects and takes the model ranked second or fifth within each subset as the target. For the rank-2 target K/n falls with n under every rule, and for the rank-5 target the mean and Borda rules keep touching about 0.4 of the tasks while the median, minimum and plurality fall. For the rank-5 target the mean rule's L_1/n stays roughly constant in n while that of the median, minimum and plurality falls; for the rank-2 target all rules fall. H3 (that K/n of the median and majority rules does not shrink) is therefore not supported for the median. The bands are wide (60 subset draws), so we make no claim about differences between neighbouring n. OLLB, with $n = 6$, is a single small-n point and supports no scaling claim.

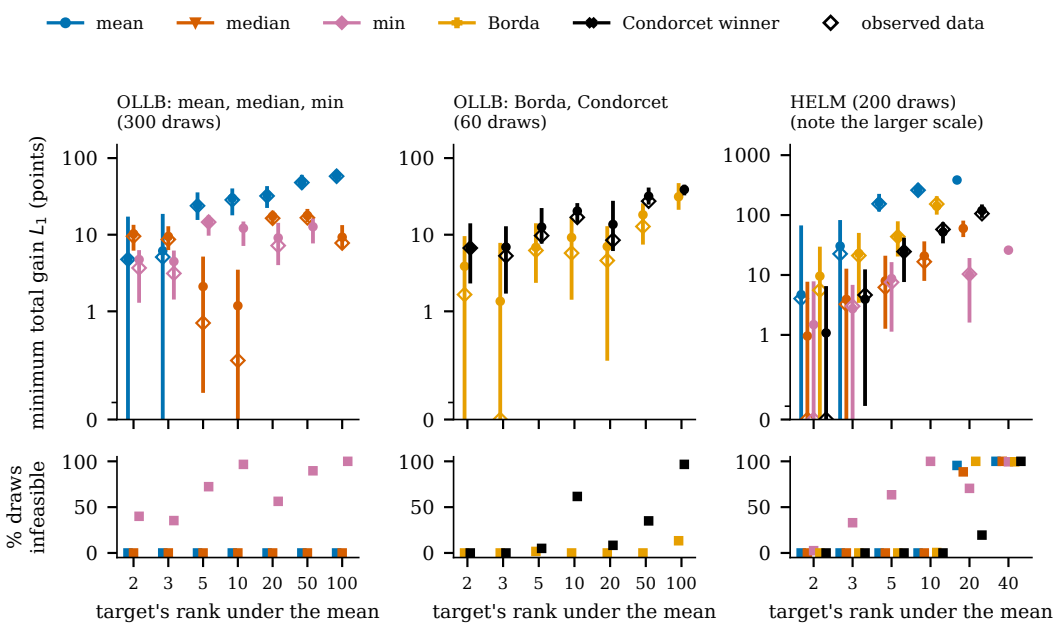


Figure 3: Stability of the minimum total gain L_1 (top, log scale) and percentage of resampled leaderboards in which the target cannot become first (bottom) under evaluation noise, for targets of the given rank under the mean: OLLB score rules (300 draws), OLLB Borda and Condorcet (60 draws; targets of ranks 10–100 are outside the 50 best targets of the frontier analysis and were solved exactly for these seven targets only) and HELM, all five rules (200 draws; larger scale). Dots and bars: median and 2.5–97.5 percentile of the draws among feasible draws (not a confidence interval of the median); open diamonds: value on the observed data, absent where the target is infeasible on it. Caps `emp99` (OLLB) and `emp99T` (HELM). Generated by `figures/src/fig_robust.py`.

Sensitivity to choices of the analysis. Appendix C (Table 6) varies the margin for "first" (1 or 2 points), merged models, raw instead of normalised OLLB scores, the 499 best OLLB targets with the rank rules, random task weights and the cost unit. The cost ordering (mean most expensive, order statistics cheapest) is unchanged in every variant that compares rules, and a 2-point margin lowers the feasible share under the mean from 12% to 10% (OLLB) and from 19% to 12% (HELM) while the median falls from 10% to 8% and from 33% to 21%. The ordering of rules by feasibility is *not* invariant: on raw OLLB scores with a +10-point cap the mean is feasible for 6.6% of targets and the median for 0.5%, against 2.5% and 4.5% on normalised scores.

Robustness (Q5). Figure 3 resamples evaluation noise (Gaussian draws with the binomial standard error of each score from the item counts; 300 draws for the OLLB score rules, 60 for OLLB Borda and Condorcet, 200 for HELM) for seven OLLB and six HELM targets; the task bootstrap (same caps, same numbers of draws) is Appendix Figure 6. The observed L_1 lies inside the 2.5–97.5 percentile interval of the draws in every case where the target is feasible on the observed data and in at least one draw (51 of 51 for noise and for the bootstrap), and below the draw median in 44 (noise) and 41 (bootstrap) of them, a small downward offset; at ranks 2–3 the lower end is 0 because some draws already make the target first. The minimum rule often makes the target infeasible in resampled leaderboards. Feasibility close to the frontier is not stable: the HELM rank-20 target, infeasible on the observed data, is feasible in 4.5% of the noise draws under the mean and 11.5% under the median, while under the minimum and the Condorcet requirement, feasible on the observed data, it is infeasible in 70.5% and 19.5% of the draws. The HELM rank-5 target needs a median of 162 points under the mean (95% interval 114–226) against 8 under the median (1–21). On OLLB the Condorcet requirement is infeasible in 0–5% of the noise draws for ranks 2, 3 and 5 but in 8%, 62% and 35% for ranks 20, 10 and 50, and in 22–63% of the task bootstraps. A target whose gap to first place is within the evaluation noise therefore cannot be said to need any particular investment: a win by the margin ε would not survive resampling.

6 Discussion and limitations

What the results say. Under caps taken from observed fine-tuning gains, a developer who can lift only the target's own scores, and whose rivals stand still, tops these leaderboards only from near the top, and how near depends on the rule: the median is feasible for more targets than the mean on HELM but for fewer on OLLB (though it reaches isolated targets further down), and the minimum is rarely feasible under the data-grounded caps. Where it is feasible the mean rule demands the most points, because it must close n times the gap (Proposition 1) whatever the spread, while the median and minimum pay only for a majority of tasks or for the weak tasks. Among the score rules the mean is thus the most expensive to manipulate in points; whether it is also the hardest to reach first place under depends on the cap and the dataset (the median is cheap in points and, on HELM, feasible for more targets, while on OLLB the mean is feasible for slightly more). The minimum is cheap when feasible but needs every weak task to be raised, so it is limited by the cap under tight caps and the most feasible rule under loose uniform caps on OLLB. Borda and the Condorcet requirement lie between the mean and the median in points. These orderings come with caveats.

Conditions under which the conclusions fail or are not established. (1) *The cap.* Feasibility is determined by the cap, not by the rule, far from the top (oracle rows of Table 1). The OLLB cap uses gains of published fine-tunes over their declared base models; this is selection-biased toward successes and says nothing about what a determined developer could achieve with a new data source. The HELM cap is a transfer of the OLLB MMLU-PRO gain ($+7.06$ points) to 57 subjects and has not been measured on HELM; the HELM frontier is set by it, since the pooled six-task cap ($+44.9$ points) makes 98.7% of HELM targets feasible under the mean. Headline claims are therefore stated per dataset, and the ordering of rules by feasibility is shown to reverse across caps. (2) *Independence across tasks.* We let each task be raised at no cost to others; targeted training can trade off tasks, which would make the true cost higher. (3) *Fixed rivals.* Other developers do not respond; leaderboards with many strategic developers need a game model such as that of Chen et al. [2026]. (4) *Score tables, not training.* The investment is in score points, not compute, data or money, so "cheap" means few score points, not little effort. (5) *Noise.* The margin of 10^{-3} points that defines "first" is below the evaluation noise of the scores and would not survive resampling (Figure 3). (6) *Rule coverage and size.* Rank rules were solved exactly for the 50 best OLLB targets only, so their feasibility fractions there cannot be compared with those of the score rules; the Condorcet requirement is a sufficient condition for Condorcet-consistent rules and therefore an upper bound for Copeland-type rules; Kemeny and Schulze were not studied. (7) *Small n and M.* OLLB has six tasks and HELM 78 targets, so HELM rank bins are small and their intervals wide; scaling in n comes from HELM subsets only. (8) *Snapshots.* Both leaderboards are frozen (OLLB 2025-03, HELM v1.13.0); nothing is claimed about later models. (9) *Not run or unfinished.* The planned exploratory overlap of optimal task sets across rules (Q6) was not run, and the comparison with observed headroom (Q7) is covered only by the feasible shares under `emp99`. Rank rules on OLLB are solved exactly only for the 50 best targets under the primary and oracle caps (499 under the mean for the slice of Appendix C); Copeland on OLLB, and rank rules under the uniform and percentile caps other than `emp99` and `ceil` on OLLB, are not solved.

Broader impact. The analysis tells leaderboard designers how exposed each rule is to targeted training under stated assumptions, which supports choosing rules and caps (for instance holding out sub-datasets) deliberately. It could also be read as a recipe for gaming a ranking. We think the risk is small: it needs no capability beyond fine-tuning, and for most models first place is infeasible under the observed gains.

7 Reproducibility

All numbers, tables and figures are produced by the supplementary code from the pinned public data (versions, checksums, seeds, commands, run times and the map from each result file to its script are in `REPRODUCE.md`; the map from each figure and table to its script and inputs is in `README.md`). The solvers are deterministic; random draws use seed 0. The pipeline needs no GPU and takes a few CPU hours on two shared CPUs. Neither leaderboard declares a licence for its score data; we release only derived tables. No material about the published study behind this question was searched for, read or encountered.

References

Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia, editors. *Handbook of Computational Social Choice*. Cambridge University Press, 2016. doi: 10.1017/CBO9781107446984.

Yatong Chen, Guanhua Zhang, and Moritz Hardt. Leaderboard incentives: Model rankings under strategic post-training, 2026. arXiv:2603.08371.

Pierre Colombo, Nathan Noiry, Ekhine Irurozki, and Stephan Clemencon. What are the best systems? new perspectives on nlp benchmarking, 2022. arXiv:2202.03799.

Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery, 2021. arXiv:2107.07002.

Piotr Faliszewski, Edith Hemaspaandra, and Lane A. Hemaspaandra. How hard is bribery in elections? *Journal of Artificial Intelligence Research*, 35:485–532, 2009. doi: 10.1613/jair.2676.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2020. arXiv:2009.03300.

Marc Lanctot, Kate Larson, Yoram Bachrach, Luke Marris, Zun Li, Avishkar Bhoopchand, Thomas Anthony, Brian Tanner, and Anna Koop. Evaluating agents using social choice theory, 2023. arXiv:2312.03121.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue Wang, Keshav Santhanam, Laurel Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan Kim, Neel Guha, Niladri Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Koreeda. Holistic evaluation of language models, 2022. arXiv:2211.09110.

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. The leaderboard illusion, 2025. arXiv:2504.20879.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. Challenging big-bench tasks and whether chain-of-thought can solve them, 2022. arXiv:2210.09261.

Shuo Yang, Wei-Lin Chiang, Lianmin Zheng, Joseph E. Gonzalez, and Ion Stoica. Rethinking benchmark and contamination for language models with rephrased samples, 2023. arXiv:2311.04850.

Kun Zhou, Yutao Zhu, Zhipeng Chen, Wentong Chen, Wayne Xin Zhao, Xu Chen, Yankai Lin, Ji-Rong Wen, and Jiawei Han. Don't make your llm an evaluation benchmark cheater, 2023. arXiv:2311.01964.

A Proofs of Propositions 1–3

Proposition 1. $f(y) = \frac{1}{n}\sum_t y_t \geq \tau$ iff $\sum_t (y_t - x_t) \geq n(\tau - f(x))$, and $\sum_t (y_t - x_t) \leq \sum_t h_t$ with each $0 \leq y_t - x_t \leq h_t$; so feasibility is $\sum_t h_t \geq n(\tau - f(x))$, i.e. $\sum_t c_t \geq n\tau$, and the minimum of the (linear) total is the requirement itself, attained by any feasible spread. K is minimised by filling the largest headrooms first, since touching k tasks can supply at most the sum of the k largest h_t. The uniform raise by $\tau - f(x)$ attains L_∞, and no spread has a smaller maximum because the average raise is $\tau - f(x)$.

Proposition 2. $\min_t y_t > \tau$ requires $y_t > \tau$ for every t, hence $c_t > \tau$ wherever $x_t \leq \tau$ (tasks with $x_t > \tau$ need no change). The cheapest y_t on a deficient task is τ (up to ε), so $L_1 = \sum_t (\tau - x_t)^+$, and K counts the deficient tasks. Both are forced.

Proposition 3. For odd n the median exceeds τ iff at least m coordinates exceed τ. Tasks with $x_t > \tau$ already count. We need $m' = (m - \#\{x_t > \tau\})^+$ more tasks. Lifting task t above τ costs at least d_t (and is possible iff $c_t > \tau$); costs are separable, so the cheapest set is the m' smallest finite d_t and $K = m'$, which cannot be smaller because each lifted task adds one to the count.

B Further results

Table 3: Closed forms (Propositions 1–3) against the exact solvers and an independent generic LP/MILP (60-target sample), over all caps. Generated by `figures/src/tables.py` from `results/theory_check.csv`.

data	rule	reference	max$\lvert\Delta L_1\rvert$	max$\lvert\Delta K\rvert$	feas. mismatches	caps
OLLB	mean	exact solver	5.7e-14	0	0	7
OLLB	mean	generic LP/MILP	8.5e-14	0	0	7
OLLB	min	exact solver	0.0e+00	0	0	7
OLLB	min	generic LP/MILP	4.3e-14	0	0	7
OLLB	median	exact solver	solver only	–	–	7
HELM	mean	exact solver	4.5e-13	0	0	8
HELM	mean	generic LP/MILP	1.8e-12	0	0	8
HELM	min	exact solver	0.0e+00	0	0	8
HELM	min	generic LP/MILP	1.3e-12	0	0	8
HELM	median	exact solver	5.7e-14	0	0	8
HELM	median	generic LP/MILP	1.5e-12	0	0	8

Table 4: Feasibility frontier of Table 1 with the analysis' 95% bootstrap intervals over targets for r^* and for "all feasible up to" (intervals of a maximum or a prefix are one-sided by construction and describe which targets are in the sample, not noise in the scores). Generated by `figures/src/tables.py` from `results/summary/frontier_*.csv`.

cap	rule	targets	feasible	share	r^* [95% CI]	all feas. up to [95% CI]
primary (`emp99`)	mean	4495	546	0.12	901 [879, 901]	394 [394, 430]
primary (`emp99`)	median	4495	451	0.10	1207 [1145, 1207]	90 [90, 133]
primary (`emp99`)	min	4495	273	0.06	2225 [1360, 2225]	5 [5, 9]
primary (`emp99`)	Borda†	49	49	1.00	50 [47, 50]	50 [47, 50]
primary (`emp99`)	Condorcet†	49	42	0.86	50 [47, 50]	18 [18, 26]
oracle (`ceil`)	mean	4495	4495	1.00	4496 [4493, 4496]	4496 [4493, 4496]
oracle (`ceil`)	median	4495	4495	1.00	4496 [4493, 4496]	4496 [4493, 4496]
oracle (`ceil`)	min	4495	4495	1.00	4496 [4493, 4496]	4496 [4493, 4496]
oracle (`ceil`)	Borda†	49	49	1.00	50 [47, 50]	50 [47, 50]
oracle (`ceil`)	Condorcet†	49	49	1.00	50 [47, 50]	50 [47, 50]

cap	rule	targets	feasible	share	r^* [95% CI]	all feas. up to [95% CI]
primary (`emp99T`)	mean	78	15	0.19	16 [13, 16]	16 [16, 19]
primary (`emp99T`)	median	78	26	0.33	29 [25, 29]	19 [19, 30]
primary (`emp99T`)	min	78	9	0.12	20 [9, 20]	6 [6, 11]
primary (`emp99T`)	Borda	78	16	0.21	17 [14, 17]	17 [17, 20]
primary (`emp99T`)	Condorcet	78	25	0.32	26 [23, 26]	26 [26, 30]
primary (`emp99T`)	Copeland	78	25	0.32	26 [23, 26]	26 [26, 29]
oracle (`ceil`)	mean	78	78	1.00	79 [76, 79]	79 [76, 79]
oracle (`ceil`)	median	78	78	1.00	79 [76, 79]	79 [76, 79]
oracle (`ceil`)	min	78	78	1.00	79 [76, 79]	79 [75, 79]
oracle (`ceil`)	Borda	78	78	1.00	79 [76, 79]	79 [76, 79]
oracle (`ceil`)	Condorcet	78	78	1.00	79 [76, 79]	79 [76, 79]
oracle (`ceil`)	Copeland	78	78	1.00	79 [76, 79]	79 [76, 79]

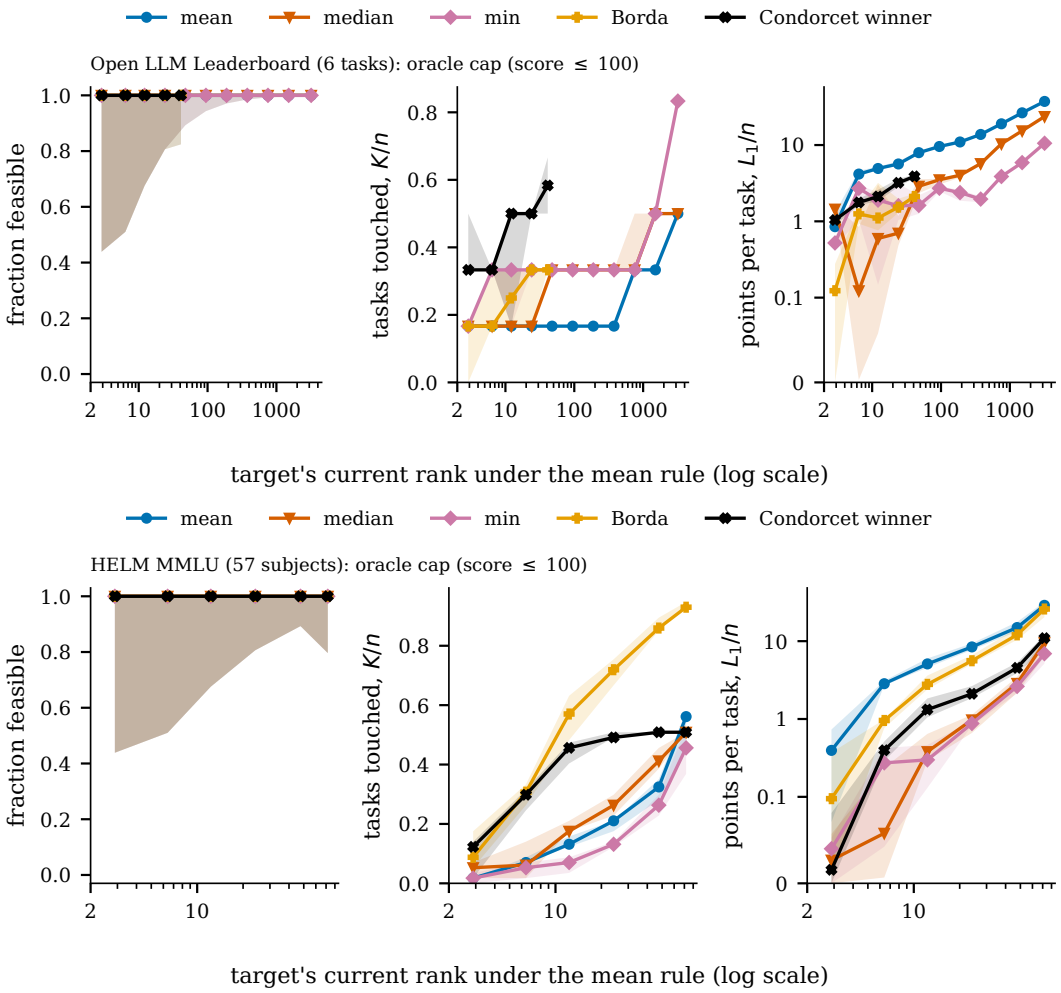


Figure 4: As Figure 1 with the oracle cap `ceil` (scores up to 100): a lower bound on any required investment; every target is feasible, so the differences between rules are not caused by the cap.

C Sensitivity variants

All entries are medians among feasible targets with the feasible share, from `results/summary/summary_variants.csv` (produced by `code/make_tables.sh`); baselines are the primary-cap rows of the same dataset. "Ratio" is the median ratio of L_1 to the baseline's on targets feasible in both.

Margin. Requiring the target to lead by 1 or 2 points instead of 10^{-3}: feasible share on OLLB (`emp99`) for mean / median / minimum is 12.1/10.0/6.1% at the baseline, 11.1/9.0/4.3% at 1 point and 10.4/7.6/2.6% at 2 points; on HELM (`emp99T`) 19/33/12% at the baseline, 15/28/12% and 12/21/9%. The median L_1 on jointly feasible targets rises by factors 1.1/1.2/1.2 (OLLB, 1 point) and 1.2/1.3/1.5 (2 points), and on HELM by 1.3/1.5/1.5 and 1.7/2.3/2.3, so the order-statistic rules are more sensitive to the margin than the mean, since a margin has to be bought on several tasks.

OLLB slices. Without merged models (3788 targets), `emp99`: feasible 11.5% (mean), 9.4% (median), 7.0% (minimum), against 12.1/10.0/6.1% on the full data; costs are unchanged for the mean and median (ratio 1.00) and 0.81 for the minimum. On raw (unnormalised) scores the oracle-cap cost of the mean is 154 points and that of the median 93, and with a $+10$-point cap the feasible shares are 6.6/0.5/5.0% (mean/median/minimum) against 2.5/4.5/9.9% on normalised scores: the ordering of rules by feasibility is not invariant to the score normalisation (the cap is in the units of the scores).

Table 5: Rule versus mean on targets feasible under both (primary caps): $n_{\text{both}}/n_{\text{either}}$, median ratio of cost(rule) to cost(mean) with 95% bootstrap interval over targets, and share of jointly feasible targets on which the rule is cheaper. A ratio below 1 means the rule is cheaper; the comparison is restricted to a selected subset (see the first column). †: the 50 best targets only. Generated by `figures/src/tables.py` from `results/summary/paired_*.csv`.

rank stratum	rule vs mean	$n_{\text{both}}/n_{\text{either}}$	L_1 ratio [95% CI]	rule cheaper (L_1)	K ratio [95% CI]	rule needs fewer tasks
all	median	435/561	0.39 [0.38, 0.40]	0.99	0.75 [0.75, 0.75]	0.56
	min	237/581	0.11 [0.10, 0.11]	1.00	0.50 [0.50, 0.67]	0.91
	Borda†	48/546	0.27 [0.25, 0.30]	1.00	1.00 [0.67, 1.00]	0.38
	Condorcet†	42/546	0.52 [0.47, 0.57]	0.95	2.00 [1.67, 2.00]	0.05
2-5	median	4/4	1.38 [0.03, 2.00]	0.25	1.00 [1.00, 1.00]	0.00
	min	4/4	0.61 [0.35, 0.77]	1.00	1.00 [1.00, 2.00]	0.00
	Borda†	3/4	0.26 [0.09, 0.35]	1.00	1.00 [1.00, 1.00]	0.00
	Condorcet†	4/4	0.91 [0.41, 1.41]	0.50	2.50 [2.00, 3.00]	0.00
6-10	median	4/5	0.02 [0.00, 0.18]	1.00	0.75 [0.50, 1.00]	0.50
	min	1/5	–	–	–	–
	Borda†	5/5	0.20 [0.18, 0.36]	1.00	1.00 [1.00, 1.00]	0.00
	Condorcet†	5/5	0.59 [0.40, 0.59]	1.00	2.00 [1.50, 3.00]	0.00
11-30	median	20/20	0.13 [0.09, 0.29]	1.00	0.50 [0.50, 0.50]	0.85
	min	12/20	0.21 [0.10, 0.28]	1.00	1.00 [0.50, 1.00]	0.33
	Borda†	20/20	0.27 [0.23, 0.37]	1.00	1.00 [0.50, 1.00]	0.45
	Condorcet†	16/20	0.51 [0.37, 0.56]	1.00	2.00 [1.50, 2.00]	0.12
31-100	median	69/70	0.37 [0.32, 0.40]	1.00	0.75 [0.67, 1.00]	0.58
	min	29/70	0.14 [0.10, 0.18]	1.00	0.67 [0.50, 0.67]	0.93
	Borda†	20/70	0.27 [0.25, 0.31]	1.00	1.00 [0.67, 1.00]	0.45
	Condorcet†	17/70	0.49 [0.47, 0.57]	1.00	1.67 [1.50, 2.00]	0.00

rank stratum	rule vs mean	$n_{\text{both}}/n_{\text{either}}$	L_1 ratio [95% CI]	rule cheaper (L_1)	K ratio [95% CI]	rule needs fewer tasks
all	median	15/26	0.07 [0.04, 0.10]	1.00	0.27 [0.23, 0.31]	0.87
	min	7/16	0.05 [0.02, 0.06]	1.00	0.12 [0.05, 0.25]	1.00
	Borda	15/16	0.51 [0.32, 0.59]	0.93	0.92 [0.90, 0.94]	0.80
	Condorcet	15/25	0.21 [0.12, 0.28]	1.00	0.67 [0.59, 0.82]	0.80
	Copeland	15/25	0.21 [0.12, 0.28]	1.00	0.67 [0.59, 0.82]	0.80
6-10	median	5/5	0.06 [0.00, 0.09]	1.00	0.27 [0.04, 0.35]	1.00
	min	2/5	–	–	–	–
	Borda	5/5	0.36 [0.32, 0.59]	1.00	0.92 [0.87, 1.00]	0.80
	Condorcet	5/5	0.19 [0.09, 0.28]	1.00	0.67 [0.54, 0.78]	1.00
	Copeland	5/5	0.19 [0.09, 0.28]	1.00	0.67 [0.54, 0.78]	1.00
11-20	median	6/9	0.09 [0.07, 0.11]	1.00	0.25 [0.22, 0.27]	1.00
	min	2/7	–	–	–	–
	Borda	6/7	0.55 [0.51, 0.64]	1.00	0.92 [0.91, 0.94]	1.00
	Condorcet	6/10	0.28 [0.21, 0.31]	1.00	0.59 [0.52, 0.64]	1.00
	Copeland	6/10	0.28 [0.21, 0.31]	1.00	0.59 [0.52, 0.64]	1.00

For the 499 best targets under the mean (`emp99`) the feasible shares are 95% (mean), 91% (Borda), 79% (median), 56% (plurality), 40% (minimum) and 30% (Condorcet requirement), and the median L_1 among feasible targets is 70, 37, 25, 29, 7.6 and 42 points respectively; one Borda instance hit the time limit and is counted as censored.

Weights. With random positive task weights (500 draws per target; weights Dirichlet with concentration 1 or 5; targets of ranks 2, 3, 5, 10, 20, 50, 100 on OLLB and 2, 3, 5, 10, 20, 40 on HELM) the weighted mean is cheaper than the equal-weight mean by a median factor 0.54 (concentration 1) and 0.70 (5) on OLLB and 0.43 and 0.68 on HELM; with all weight on one task (OLLB, 42 target–task pairs) the ratio is 0.38. Feasible shares are 91% and 99% on OLLB and 68% and 67% on HELM (equal weights: 100% and 67%).

Cost in standard-deviation units. Measuring each task's raise in units of the task's standard deviation across models, the median total cost under `emp99` is 4.3 (mean) and 1.6 (minimum) on OLLB, and 17.5 (mean), 2.5 (median) and 0.6 (minimum) on HELM, so the ordering of costs is the same as in points.

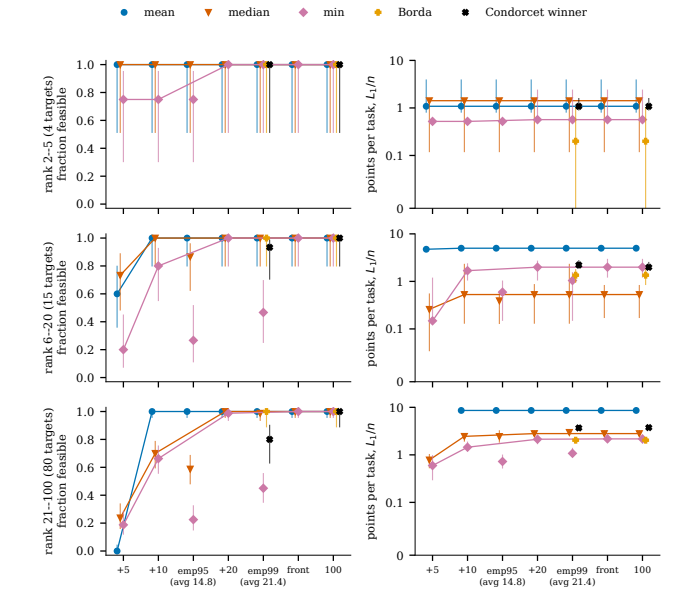
Table 6: Sensitivity of the primary results (OLLB, cap `emp99`; HELM, `emp99T`): share of targets that can reach first place under the variant and under the baseline, median L_1 among feasible targets, and median ratio of L_1 to the baseline on targets feasible under both (paired n). For random weights n counts targets $\times$ weight draws at stratified ranks, so shares are not population fractions. Generated by `figures/src/tables.py` from `results/summary/summary_variants.csv`.

variant	cap	rule	n	feasible: variant / base	median L_1	median K	L_1 ratio to base (paired n)	cens.
margin 1 pt	emp99	mean	4495	0.11 / 0.12	77.1	4	1.08 (500)	0
margin 1 pt	emp99	median	4495	0.09 / 0.10	29.5	3	1.15 (403)	0
margin 1 pt	emp99	min	4495	0.04 / 0.06	9.6	2	1.25 (191)	0
margin 2 pt	emp99	mean	4495	0.10 / 0.12	82.0	4	1.17 (466)	0
margin 2 pt	emp99	median	4495	0.08 / 0.10	32.2	3	1.34 (341)	0
margin 2 pt	emp99	min	4495	0.03 / 0.06	10.9	2	1.53 (115)	0
merged models dropped	ceil	mean	3788	1.00 / 1.00	183.3	2	1.00 (3788)	0
merged models dropped	ceil	median	3788	1.00 / 1.00	108.0	3	1.00 (3787)	0
merged models dropped	ceil	min	3788	1.00 / 1.00	42.4	3	0.93 (3787)	0
merged models dropped	emp99	mean	3788	0.11 / 0.11	74.1	4	1.00 (435)	0
merged models dropped	emp99	median	3788	0.09 / 0.09	26.6	3	1.00 (356)	0
merged models dropped	emp99	min	3788	0.07 / 0.06	7.4	2	0.81 (212)	0
raw scores	ceil	mean	4495	1.00 / 1.00	154.1	2	0.85 (4495)	0
raw scores	ceil	median	4495	1.00 / 1.00	92.9	3	0.86 (4494)	0
raw scores	ceil	min	4495	1.00 / 1.00	74.7	5	1.61 (4493)	0
raw scores	gain10	mean	4495	0.07 / 0.03	48.5	5	0.74 (113)	0
raw scores	gain10	median	4495	0.00 / 0.05	15.7	2	5.65 (20)	0
raw scores	gain10	min	4495	0.05 / 0.10	12.6	2	1.50 (217)	0
500 best models only	ceil	Borda	499	1.00 / –	36.3	4	0.91 (48)	1
500 best models only	ceil	Condorcet	499	1.00 / –	47.0	4	1.00 (49)	0
500 best models only	ceil	mean	499	1.00 / 1.00	71.1	1	1.00 (499)	0
500 best models only	ceil	median	499	1.00 / 1.00	26.5	2	1.00 (498)	0
500 best models only	ceil	min	499	1.00 / 1.00	12.3	2	1.00 (498)	0
500 best models only	emp99	Borda	499	0.91 / –	36.5	4	0.91 (48)	1
500 best models only	emp99	Condorcet	499	0.30 / –	42.2	5	1.00 (42)	0
500 best models only	emp99	mean	499	0.95 / 0.95	70.3	3	1.00 (473)	0
500 best models only	emp99	median	499	0.79 / 0.79	25.5	3	1.00 (395)	0
500 best models only	emp99	min	499	0.40 / 0.40	7.6	2	1.00 (200)	0
random weights†	emp99	Dirichlet(1)	3500	0.91 / 1.00	12.2	1	0.54 (3128)	0
random weights†	emp99	Dirichlet(5)	3500	0.99 / 1.00	19.1	1	0.70 (3440)	0
random weights†	emp99	one-hot	42	0.33 / 1.00	8.5	1	0.38 (14)	0
cost in noise SE‡	ceil	mean	4495	1.00 / –	11.1	–	–	0
cost in noise SE‡	ceil	min	4495	1.00 / –	6.4	–	–	0
cost in noise SE‡	emp99	mean	4495	0.12 / –	4.3	–	–	0
cost in noise SE‡	emp99	min	4495	0.06 / –	1.6	–	–	0

variant	cap	rule	n	feasible: variant / base	median L_1	median K	L_1 ratio to base (paired n)	cens.
margin 1 pt	emp99T	Borda	78	0.19 / 0.21	116.3	31	1.16 (15)	0
margin 1 pt	emp99T	Condorcet	78	0.32 / 0.32	88.0	27	1.14 (25)	0
margin 1 pt	emp99T	mean	78	0.15 / 0.19	229.1	33	1.33 (12)	0
margin 1 pt	emp99T	median	78	0.28 / 0.33	30.1	11	1.55 (22)	0
margin 1 pt	emp99T	min	78	0.12 / 0.12	4.7	3	1.50 (8)	0
margin 2 pt	emp99T	Borda	78	0.19 / 0.21	117.2	31	1.17 (15)	0
margin 2 pt	emp99T	Condorcet	78	0.32 / 0.32	90.2	27	1.17 (25)	0
margin 2 pt	emp99T	mean	78	0.12 / 0.19	270.3	39	1.73 (9)	0
margin 2 pt	emp99T	median	78	0.21 / 0.33	38.8	12	2.34 (16)	0
margin 2 pt	emp99T	min	78	0.09 / 0.12	6.7	3	2.26 (6)	0
random weights†	emp99T	Dirichlet(1)	3000	0.68 / 0.67	33.6	6	0.43 (1739)	0
random weights†	emp99T	Dirichlet(5)	3000	0.67 / 0.67	74.9	12	0.68 (1836)	0
cost in noise SE‡	ceil	mean	78	1.00 / –	52.0	–	–	0
cost in noise SE‡	ceil	median	78	1.00 / –	10.2	–	–	0
cost in noise SE‡	ceil	min	78	1.00 / –	8.3	–	–	0
cost in noise SE‡	emp99T	mean	78	0.19 / –	17.5	–	–	0
cost in noise SE‡	emp99T	median	78	0.33 / –	2.5	–	–	0
cost in noise SE‡	emp99T	min	78	0.12 / –	0.6	–	–	0

D Hypotheses fixed before the full results

The plan fixed three hypotheses from the closed forms before the full result set was available. H1: with loose caps, score-sum rules need the same L_1 regardless of K, while order-statistic and majority rules need more tasks but possibly fewer points than the mean. H2: required investment grows with the target's initial rank; feasibility falls as caps tighten; rule ordering can change with the cap. H3: for fixed per-task caps the number of tasks that majority and median rules need does not shrink with n, while the mean rule's L_1/n depends on the density of gaps. Anything beyond these is exploratory.



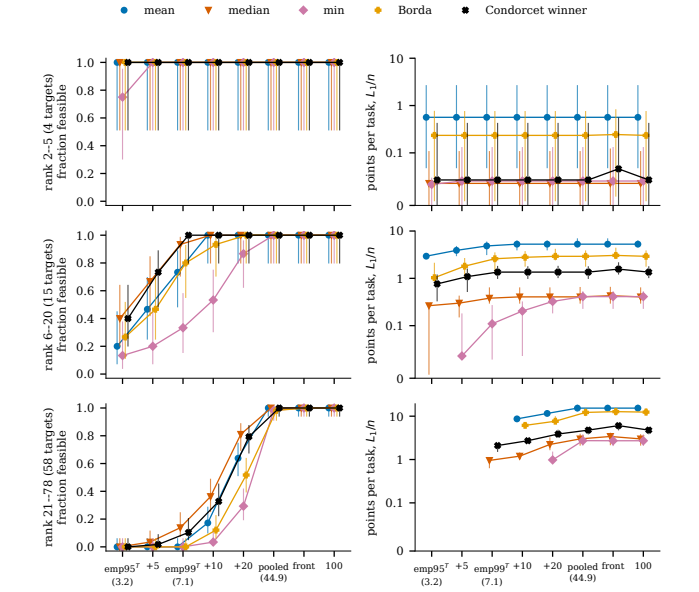


Figure 5: OLLB (top, 4495 targets) and HELM (bottom, 78 targets): effect of the cap on feasibility (left) and cost among feasible targets (right) for three rank strata, caps ordered by size in points (+h: at most h points per task; emp95/emp99: empirical percentiles of fine-tune gains, on OLLB one value per task placed at the average, on HELM the MMLU-PRO value $3.2/7.1$ used on every subject (transfer); pooled: percentile of all six tasks' gains, 44.9 points; front: best rival score; 100: ceiling). Markers are bin values with 95% intervals (Wilson for fractions, bootstrap over targets for medians). For the mean rule the cost among feasible targets does not depend on the cap (Proposition 1); only the feasible fraction does. Borda and Condorcet on OLLB (50 best targets) appear as isolated markers. Generated by `figures/src/fig_cap_grid.py`.

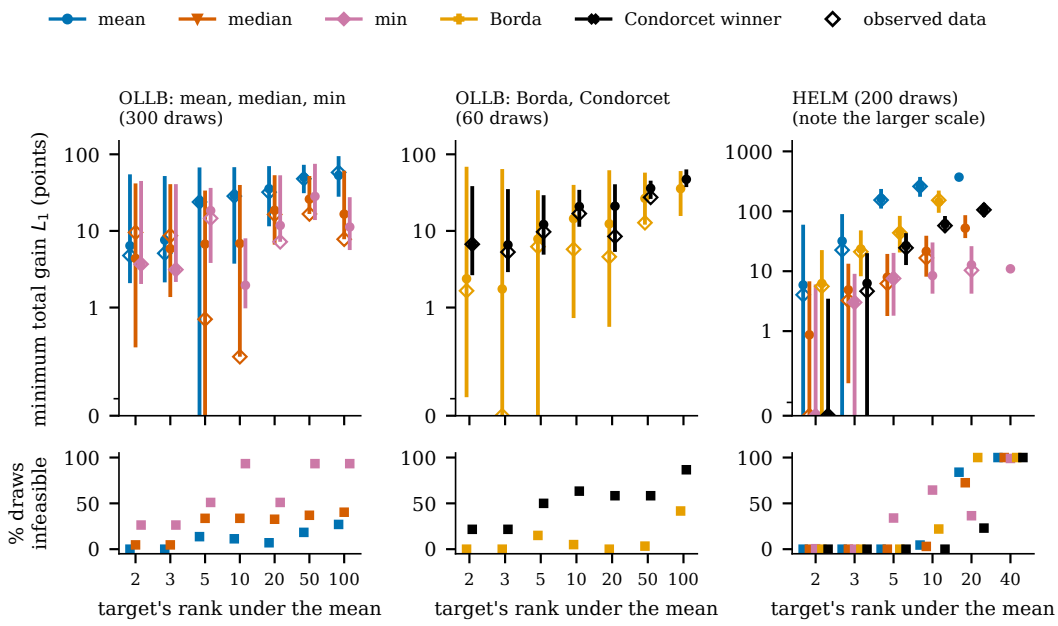


Figure 6: As Figure 3 for the bootstrap over tasks (tasks resampled with replacement, per-task caps following their task, same numbers of draws; HELM larger scale). Generated by `figures/src/fig_robust.py`.

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